

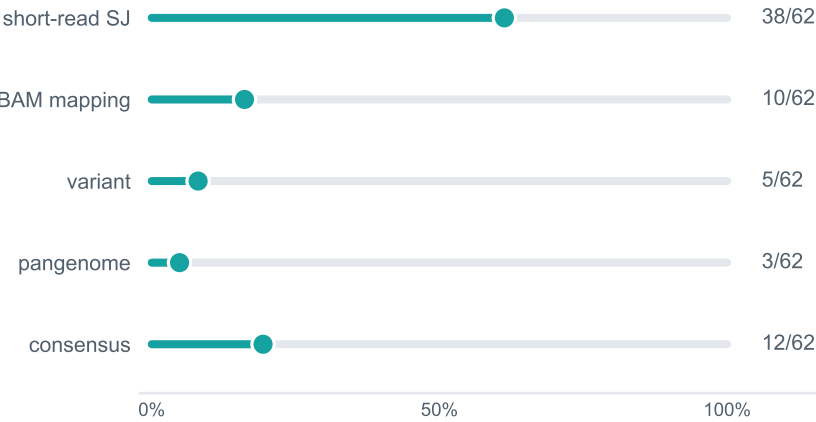
Verdict on 49 novel isoforms

Trustworthy = high/medium-confidence novel · the share you can act on at a glance



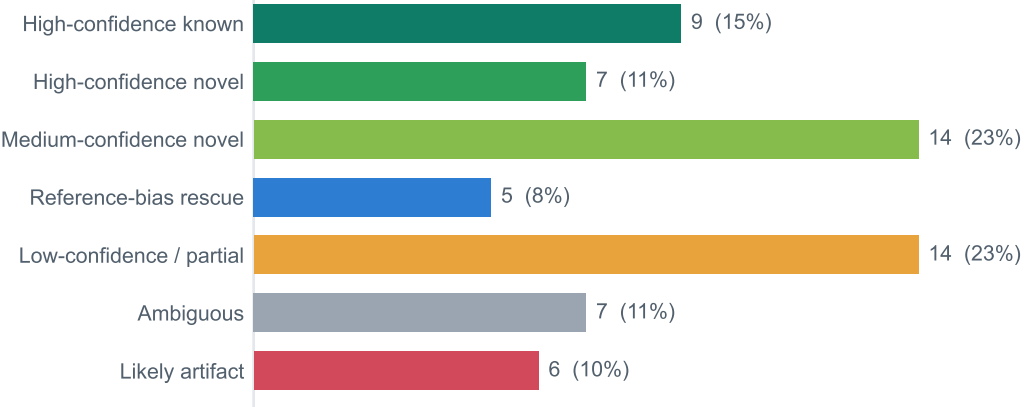
Evidence axes in play

fraction of isoforms for which each axis could be evaluated



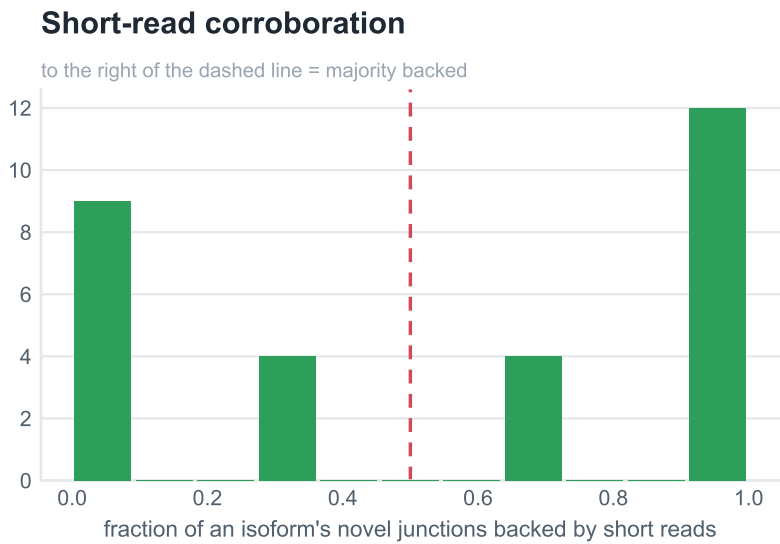
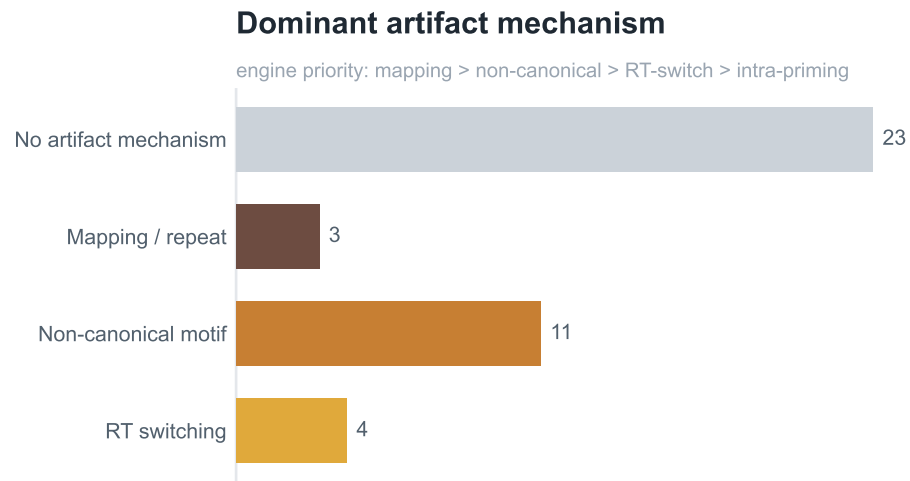
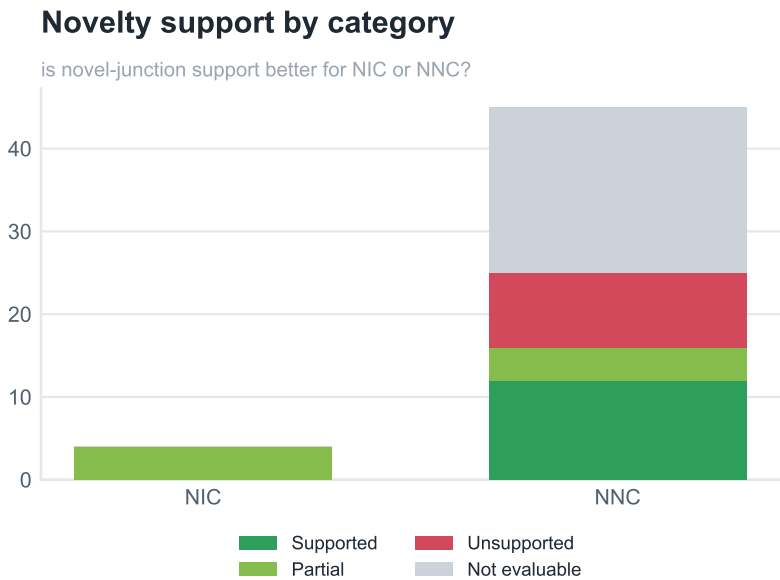
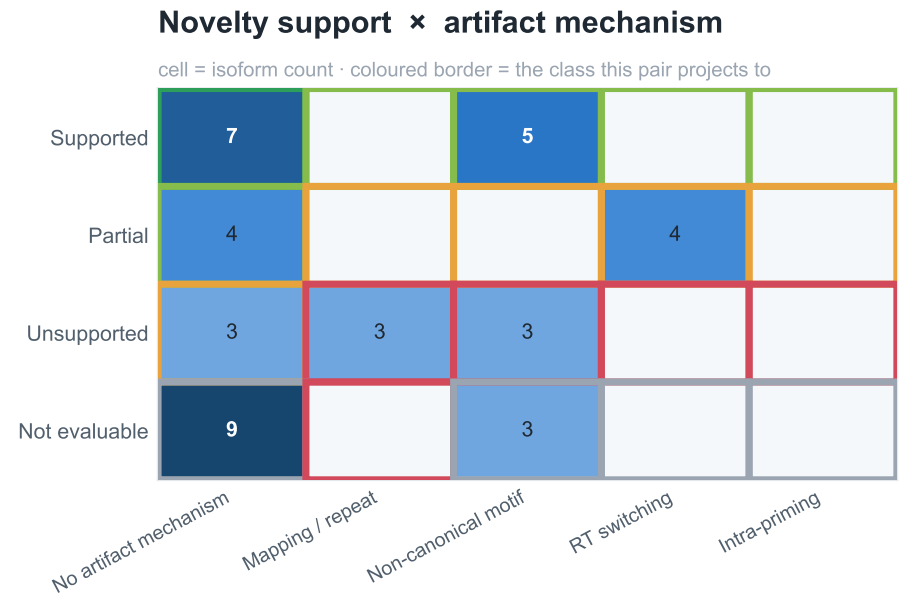
Confidence-class distribution

every isoform · count (share of all)



# How each verdict was reached

Every novel isoform is scored on two axes; the pair projects to one confidence class.

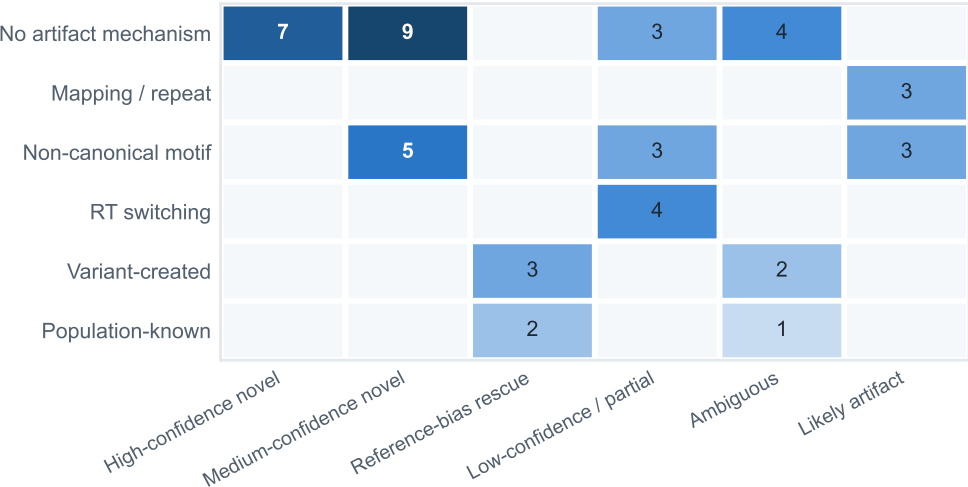


# Why calls were down-weighted

Which artifact mechanism drove each verdict, and whether the evidence supports it.

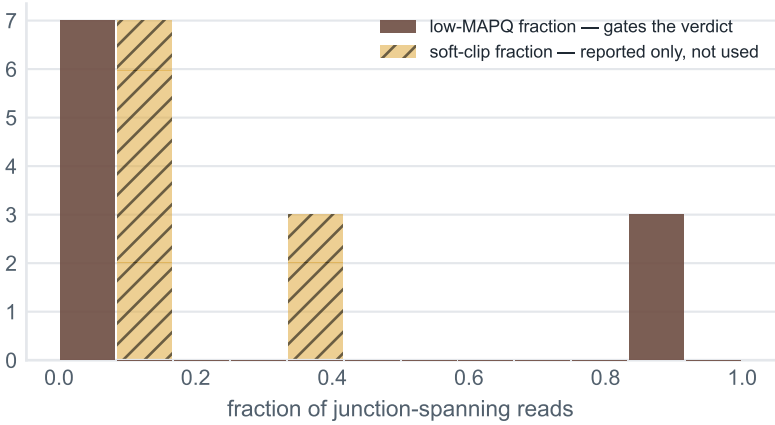
## Mechanism → resulting class

where each artifact mechanism ends up



## BAM mapping-evidence quality

are mapping artifacts backed by poor alignment?



## Structural-signal availability

null is shown as n/a — never silently counted as false

| Structural prior       | true | false | n/a |
|------------------------|------|-------|-----|
| Non-canonical motif    | 11   | 51    | 0   |
| RT-switch stage        | 4    | 58    | 0   |
| Caller chain available | 62   | 0     | 0   |

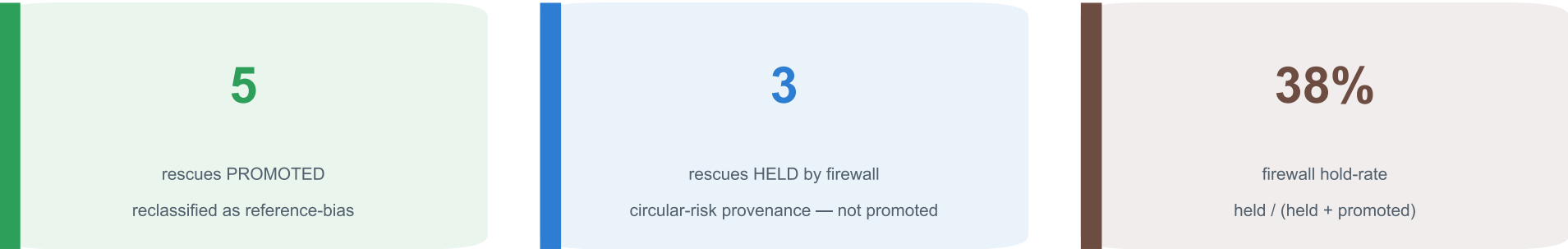
## Mapping-down-weighted calls (3)

| isoform    | class           | reads | low-MAPQ | suppl. |
|------------|-----------------|-------|----------|--------|
| NNC_map.36 | Likely artifact | 14    | 0.86     | 0.71   |
| NNC_map.37 | Likely artifact | 14    | 0.86     | 0.71   |
| NNC_map.38 | Likely artifact | 14    | 0.86     | 0.71   |

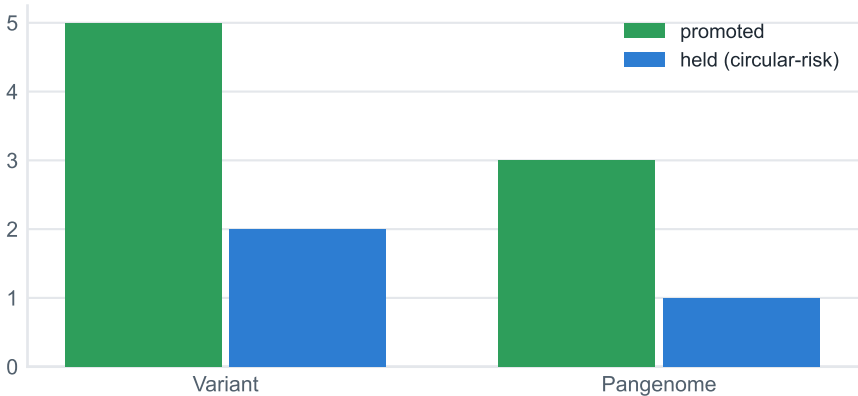
# Reference-bias rescue & circularity firewall

A novel junction explained by reference bias is reclassified — but only on independent evidence.

The thesis: nothing is promoted on circular evidence



Rescue axis: promoted vs held



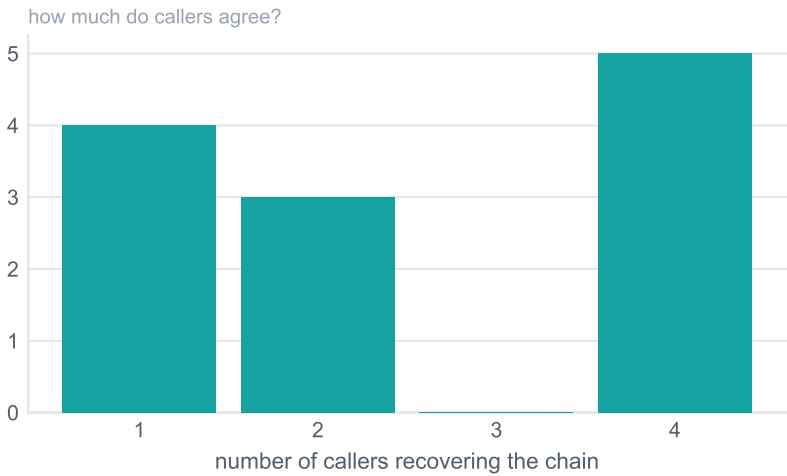
Held-rescue ledger (3)

| isoform     | category       | why held (rule trace)                                  |
|-------------|----------------|--------------------------------------------------------|
| HELD_var.44 | not_in_catalog | variant creates canonical motif on haplotype but prove |
| HELD_var.45 | not_in_catalog | variant creates canonical motif on haplotype but prove |
| HELD_pan.46 | not_in_catalog | all 2/2 novel junctions realizable on a pangenome path |

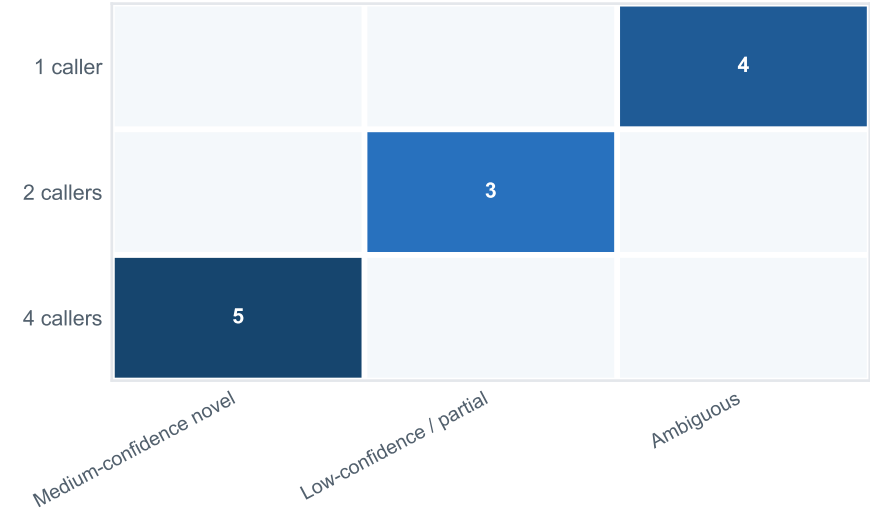
# Multi-caller consensus

Caller agreement is methodological corroboration — it promotes to MEDIUM at most, never HIGH.

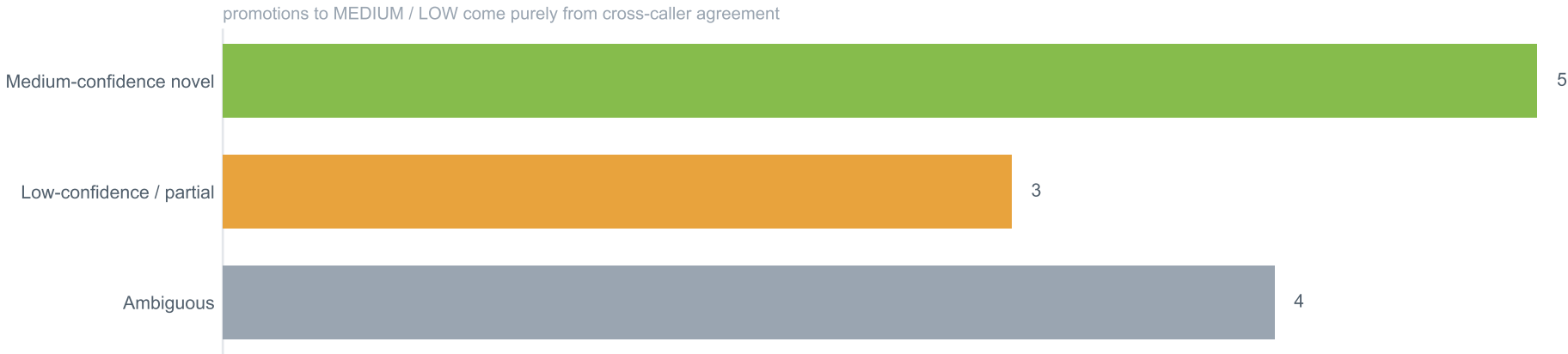
Caller-support distribution



Caller support → confidence class

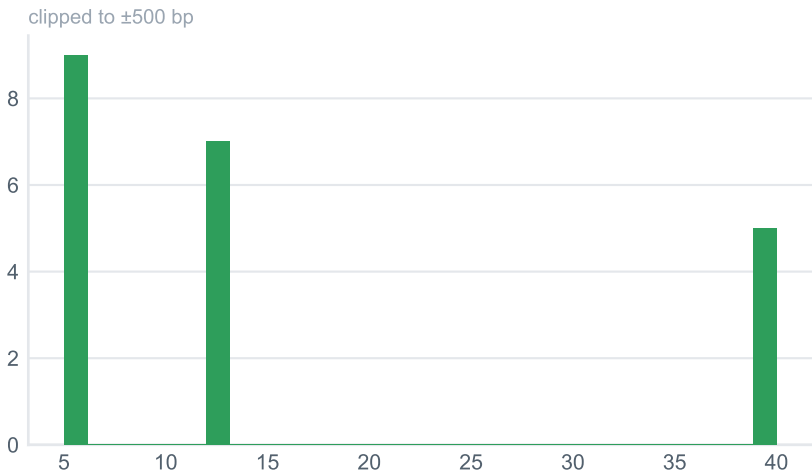


Verdicts of short-read-unknown novels under the consensus axis

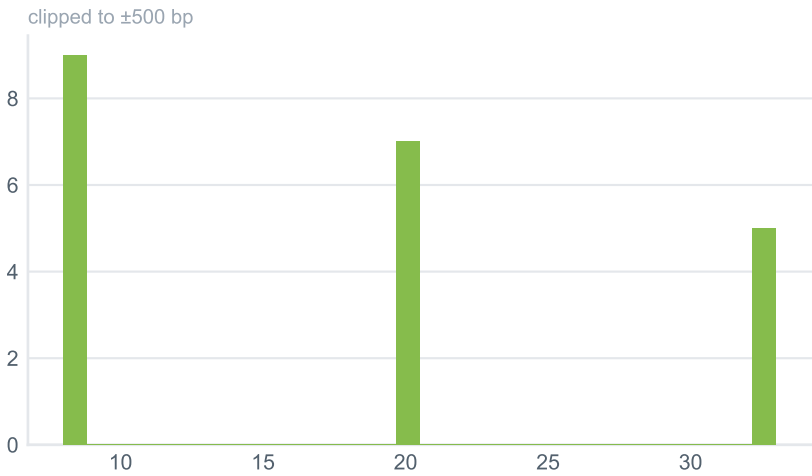


Passed through verbatim — these are NOT used in the PanIsoGuard verdict.

Distance to CAGE peak



Distance to polyA site



Predicted NMD



PolyA motif found

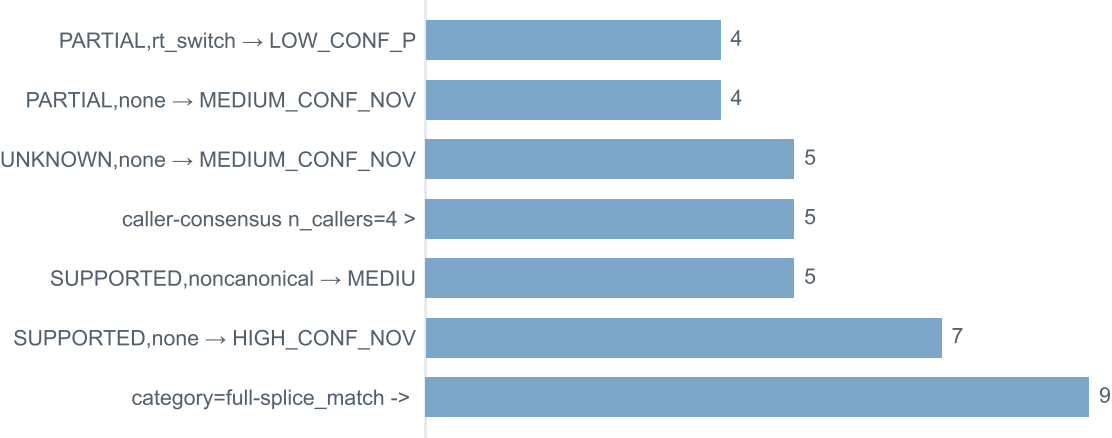


Exactly how this report was produced, and the checks that make it auditable.

Run provenance

| key                    | value                     |
|------------------------|---------------------------|
| tool_version           | 0.0.3                     |
| ruleset_version        | builtin-0.0.1             |
| sqanti3_version_target | 6.0                       |
| config                 | <built-in defaults>       |
| classification         | showcase (synthetic demo) |
| isoforms               | showcase (synthetic demo) |
| ref_gtf                | showcase (synthetic demo) |
| sj_tab                 | showcase (synthetic demo) |
| bam                    | showcase (synthetic demo) |
| axis.short_read        | on                        |
| axis.catalog           | on                        |
| axis.bam               | on                        |
| axis.variant           | on                        |
| axis.pangenome         | on                        |

Most frequent rule-trace lines



Defensibility checklist

- ✓ class tally reconciles (provenance ↔ JSONL)
- ✓ firewall held 3 circular-risk rescue(s)
- consensus axis used this run
- axes in play: short-read, BAM, variant, pangenome, consensus
- SQANTI3 priors consumed, never recomputed